

Technical Analysis of CSMA and CSMA/CD in Ethernet Networks

Summary

Carrier Sense Multiple Access (CSMA) and **CSMA with Collision Detection (CSMA/CD)** are MAC protocols developed to improve efficiency over earlier random access methods like Pure and Slotted Aloha. The key principles include:

- **CSMA:** Stations sense the medium before transmitting to reduce collision probability.
- **CSMA/CD:** Enhances CSMA by allowing stations to **detect collisions during transmission** and abort immediately, saving bandwidth.
- **Collision Detection Requirement:** Transmission delay (T_{fr}) must be at least **twice the propagation delay ($2 \times T_{p}$)** to guarantee detection.
- **Operational Efficiency:** No acknowledgments (ACKs) or frame copies are needed after transmission; successful delivery is implied if no collision occurs.

1. Evolution of Medium Access Control

Protocol	Mechanism	Vulnerable Time	Efficiency
Pure Aloha	Transmit whenever ready	$2 \times T_{fr}$	High collision rate
Slotted Aloha	Transmit only at slot start	T_{fr}	Collisions halved vs. Pure Aloha
CSMA	Listen before transmit	T_{p}	Improved; collisions still occur
CSMA/CD	Listen while transmitting; abort on collision	$2 \times T_{p}$	Maximum efficiency; reduces wasted time

Observation: CSMA/CD is the standard for Ethernet due to minimized wasted bandwidth and high efficiency.

2. Carrier Sense Multiple Access (CSMA)

Core Principles:

- **Carrier Sense (CS):** A station senses the local medium before transmitting.
- **Multiple Access (MA):** Multiple stations share the channel equally.
- **Propagation Delay Limitation:** Each station senses the medium only locally. If another station has already begun transmitting, collisions may still occur due to signal propagation delay.

Vulnerable Time:

[
 $T_{\{vul\}} = T_p$
]

Collisions occur if another station begins transmitting within this time window.

3. CSMA/CD (Collision Detection)

CSMA/CD extends CSMA by **detecting collisions in real-time**.

Dual-Port Mechanism

1. **Sending Port (SP):** Transmits data onto the medium.
2. **Listening Port (LP):** Monitors for overlapping signals (collisions) during transmission.

Key Features:

- **Simultaneous Transmission & Detection:** Both processes occur concurrently.
- **Immediate Abortion:** On collision, transmission stops, saving bandwidth.
- **No ACKs Required:** If the frame is transmitted completely without collision, delivery is guaranteed.
- **No Frame Retention:** Unlike flow-controlled protocols, CSMA/CD does not store a copy post-transmission.

4. Collision Detection Window and Jam Signals

- **Collision Detection Window:** The station must still be transmitting when the collision signal reaches it.
- **Jam Signal:** Broadcast to notify all stations of a collision.
 - Transmitting stations use it to trigger retransmission procedures.
 - Idle stations discard it.

Detection Constraint:

A collision can only be detected if:

$$\left[\begin{array}{l} T_t \geq 2 \times T_p \\ \end{array} \right]$$

or more precisely:

$$\left[\begin{array}{l} T_t \geq 2 \times T_p + T_{\text{jam}} \\ \end{array} \right]$$

5. Minimum Frame Size Calculation

Purpose: Ensure a transmitting station remains active long enough to detect collisions.

Formulas:

1. Basic:

$$\left[\begin{array}{l} T_t \geq 2 \times T_p \\ \end{array} \right]$$

2. Including Jam Signal:

$$\left[\begin{array}{l} T_t \geq 2 \times T_p + T_{\text{jam}} \\ \end{array} \right]$$

3. Frame Size (L):

$$\left[\begin{array}{l} L \geq T_t \times \text{Bandwidth} \\ \end{array} \right]$$

Expanded:

$$\left[\begin{array}{l} L \geq \left(2 \times \frac{\text{Distance}}{\text{Velocity}} + T_{\text{jam}} \right) \times \text{Bandwidth} \\ \end{array} \right]$$

6. Practical Scenarios

Network	Parameters / Scenario	Minimum Frame Size
1 Gbps	1 km, signal speed 2×10^5 km/s	10,000 bits

100 Mbps	1 km, 1250-byte requirement	1250 bytes
20 Mbps	Propagation 40 μ s	1600 bits (200 bytes)
10 Mbps	Jam signal 48 bits, RTT 46.4 μ s	512 bits
100 Mbps (GATE 2024)	50 m segment	500 bits

Observation: Frame size is directly related to propagation delay and bandwidth; higher-speed networks require larger frames to ensure collision detection.